ABESIT
Group of Institutions

HACKNOVATE 7.0

- **Problem Statement Title-** Development of AI-powered FRA Atlas and WebGIS-based Decision Support System (DSS) for Integrated Monitoring of Forest Rights Act (FRA) Implementation. (States to be concentrated: Madhya Pradesh, Tripura , Odisha, Telangana)
- **Theme- AI/ML**
- **PS Category- Software**
- **Team Name- Singularity**

PROBLEM STATEMENT AND SOLUTION

PROBLEM

- The Forest Rights Act (FRA) grants land rights to tribal and forest-dwelling communities.
- Presently, claim data (Approved, Pending, Rejected) is scattered and lacks a centralized monitoring system.

CHALLENGES

- Manual, time-consuming verification.
- Lack of transparency for citizens.
- No real-time analytics or GIS-based tracking.
- Difficulty for policymakers to make data-driven decisions.

SOLUTION - FRA Atlas

- An AI-powered, WebGIS-based Decision Support System for FRA implementation.

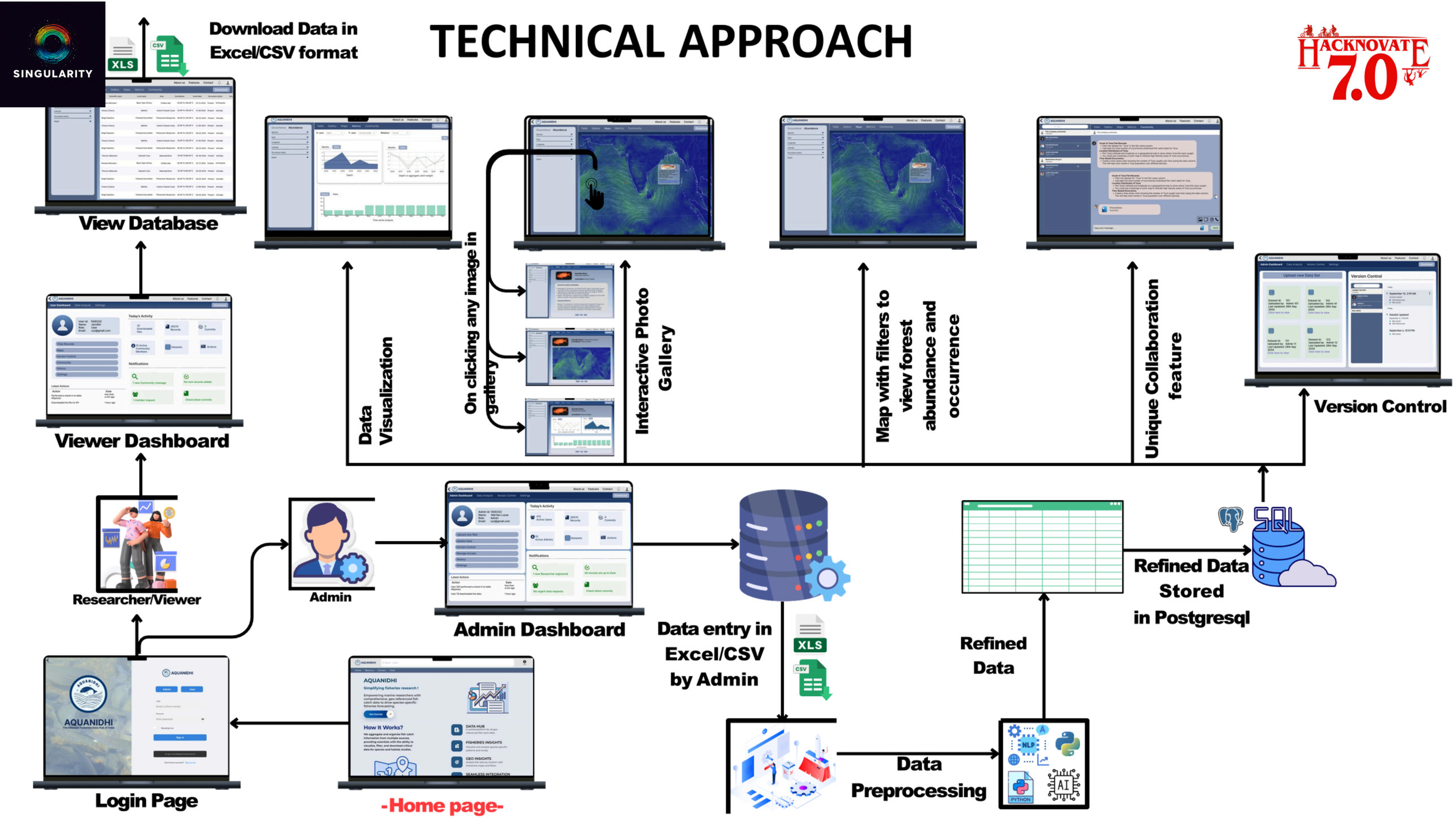
KEY FEATURES

- Centralized dashboard for claims (Approved, Pending, Rejected).
- AI-driven analytics & predictive insights.
- WebGIS maps with geo-tagged claim data.
- User roles: Govt officials (manage claims) & Citizens (track status + download reports).
- Multilingual & accessible platform.

TECH STACK

React → JS → node.js → SQLite → PostgreSQL → Leaflet → scikit learn → TensorFlow →

TECHNICAL APPROACH



FEASIBILTY AND VIABILITY

Viability

1 Practical Implementation:

- Uses available govt data and proven GIS/AI tools
- Built on existing systems, reducing cost and rollout time..

2 Adoption Potential:

- Useful for both govt officials and citizens.
- Simple, multilingual interface boosts adoption by officials and citizens.

3 Sustainability:

- Open-source tech ensures low maintenance & long-term use.
- Training local stakeholders to manage and update the system promotes self-reliance, reduces external dependency, and ensures the initiative's longevity.

4 Scalability:

- Can be expanded across states and integrated with national FRA systems.
- The system's modular design allows easy adaptation to diverse state-level policies and seamless integration with future national digital platforms.

5 Empowering Communities:

- Gives tribal communities a clear, verifiable record of their rights, enhancing security and well-being.
- With documented and verifiable rights, tribal communities can more easily access government schemes, financial services, and development programs designed for them.

6 Social Impact:

- Builds trust, transparency, and awareness among forest communities.
- By simplifying processes and improving access to information, forest communities become more empowered to participate in decision-making and local governance.

7 Policy Support:

- Provides data-driven insights for better planning and governance.
- Using accurate, real-time data helps governments direct funds and benefits to the communities most in need, ensuring more efficient and equitable use of resources.

8 Reduced Bureaucracy:

- Will automate the complex process of linking FRA rights with welfare scheme eligibility
- Automation will create a clear, trackable system for verifying FRA rights and welfare eligibility, reducing human errors and opportunities for corruption..

Feasibilty

1 Data Availability & Interoperability

- Data for FRA, satellites, and schemes is accessible, and the project's goal is to unify it.
- Combines multiple data sources into one platform for easier access and analysis.

2 Technological Maturity

- Proven technologies like OCR, GIS, and standard AI models minimize development risk and cost
- Established tools ensure stable performance and easy upgrades..

3 Scalability and Phased Implementation

- A pilot project in one region can validate the system before a full-scale, step-by-step rollout.
- Pilot testing identifies issues early, ensuring smoother expansion.

4 Strategic Alignment & User Buy-in

- The project supports key government initiatives, ensuring strong user adoption and stakeholder support.
- Aligning with government priorities builds trust and speeds up adoption.

IMPACTS AND BENEFITS

SOCIAL BENIFITS

- Promotes Social Justice: Rectifies historical injustices by securing land and resource rights for forest-dwelling communities.
- Boosts Livelihoods: Connects beneficiaries to welfare schemes and recognizes assets, leading to stable incomes and poverty reduction.
- Empowers Communities: Strengthens traditional governance, collective ownership, and cultural identity through secure CFR rights.
- Expands Essential Services: Enables targeted delivery of water, health, education, and other vital services to tribal areas.

ENVIRONMENTAL BENIFITS

- Supports Forest Conservation: Secure CFR rights motivate communities to protect and sustainably manage their traditional forest resources.
- Enhances Biodiversity: Improved community-led forest management practices contribute to the preservation of local ecosystems and species.
- Combats Climate Change: Sustainable forest use and monitoring aid in maintaining forest cover, crucial for carbon sequestration and climate resilience.
- Promotes Sustainable Resource Use: Clear mapping and transparent monitoring facilitate responsible utilization of forest and water resources, preventing degradation.

ECONOMIC BENIFITS

- Boosts Household Income: Direct scheme access and land recognition increase agricultural and forest-based earnings.
- Unlocks Credit & Investment: Documented rights facilitate access to formal credit and targeted developmental investments.
- Maximizes Resource Value: Sustainable management and clear ownership enhance the economic potential of forest resources.
- Optimizes Public Spending: Data-driven fund allocation ensures efficient use of government resources for economic development.

SHORT-TERM IMPACTS

- Enhances operational efficiency for government departments, enabling easier monitoring, reporting, and coordination across various levels.
- Patta holders begin receiving benefits earlier, leading to immediate improvements in their financial stability, access to basic services, and overall well-being.
- Quickens the pace of rights recognition, reducing uncertainty and frustration for forest-dwelling communities.

LONG-TERM IMPACTS

- Leads to more effective policy formulation, efficient resource allocation, better disaster preparedness in tribal regions, and continuous improvement in public service delivery based on empirical data.
- Contributes to biodiversity conservation, climate change mitigation through improved forest health, and equitable sharing of forest produce benefits.
- Leads to sustainable livelihood improvements, reduced poverty, enhanced food security, and greater self-reliance, fostering resilient community development.

References & Foundational Literature

Legal & Policy Framework :-

- Govt. of India (2006) - The Forest Rights Act (FRA), the project's legal basis.
- Ministry of Tribal Affairs (2022) - Annual Report on FRA progress, schemes, and official data.
- Sarin & Springate-Baginski (2010) - Seminal analysis of the FRA's legislative anatomy.
- Saxena, N. C. (2012) - Key report on on-the-ground implementation hurdles.

Technical & Implementation :-

- Agarwal & Roy (2002) - Foundational model for a WebGIS-based DSS in forestry.
- Maxwell & Warner (2020) - Comparison of AI/ML models for satellite image classification.
- Talukdar et al. (2020) - Comprehensive review of ML classifiers for land-use mapping.

Socio-Economic & Review :-

- Garg, A. (2019) - Report on challenges in India's legacy land record systems.
- Kumar & Banerjee (2021) - Systematic review of FRA implementation challenges and gaps.
- Sahoo & Dhal (2009) - Paper on applying remote sensing for tribal development planning.